



Building RAG Agents with LLMs

Product Building with Agentic Coding

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본 자료의 저작권은 Jae Y. CHOI (jaeyeong2022@gmail.com)에게 있습니다.

본 자료는 자유롭게 이용할 수 있으나, 사용 시 저작자 및 출처를 반드시 표기해 주시기 바랍니다.

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Today, You Will...

Product Building with Agentic Coding

- Reuse an existing Skill
- Build a public-data Skill
- Create and verify a useful page
- Connect MCP, deploy, and update





Building RAG Agents with LLMs

Product Building with Agentic Coding

Code Is Also a Sequence of Tokens

A tiny completion—with a checkable result

```
def add(a, b):  
    return a + b
```

```
add(2, 3)
```

5



Run the example

Producing plausible code and passing the check are different.

Code-Focused Model Families

Examples already in this course: Qwen3-Coder and DeepSeek-Coder-V2

QWEN3-CODER

Qwen3-Coder

updated Dec 31, 2025

Qwen3 Coder WebDev 🌐

Generate HTML or React code from your web app description

🔥 ❤️ 1.15k

🌟 Qwen/Qwen3-Coder-480B-A35B-Instruct

📄 Text Generation • 🗄️ 480B • Updated Aug 21, 2025 • ⬇️ 30.3k • ⚡ • ❤️ 1.37k

🌟 Qwen/Qwen3-Coder-480B-A35B-Instruct-FP8

DEEPSEEK-CODER-V2

DeepSeekCoder-V2

updated Nov 27, 2025

🌟 deepseek-ai/DeepSeek-Coder-V2-Instruct

📄 Text Generation • 🗄️ 236B • Updated Aug 21, 2024 • ⬇️ 9.65k • ❤️ 700

🌟 deepseek-ai/DeepSeek-Coder-V2-Base

📄 Text Generation • 🗄️ 236B • Updated Jul 3, 2024 • ⬇️ 2.83k • ❤️ 82

🌟 deepseek-ai/DeepSeek-Coder-V2-Lite-Base

📄 Text Generation • 🗄️ 16B • Updated Jul 3, 2024 • ⬇️ 8.36k • ❤️ 118

Code-Model Ecosystem

Model family overview · not a current leaderboard

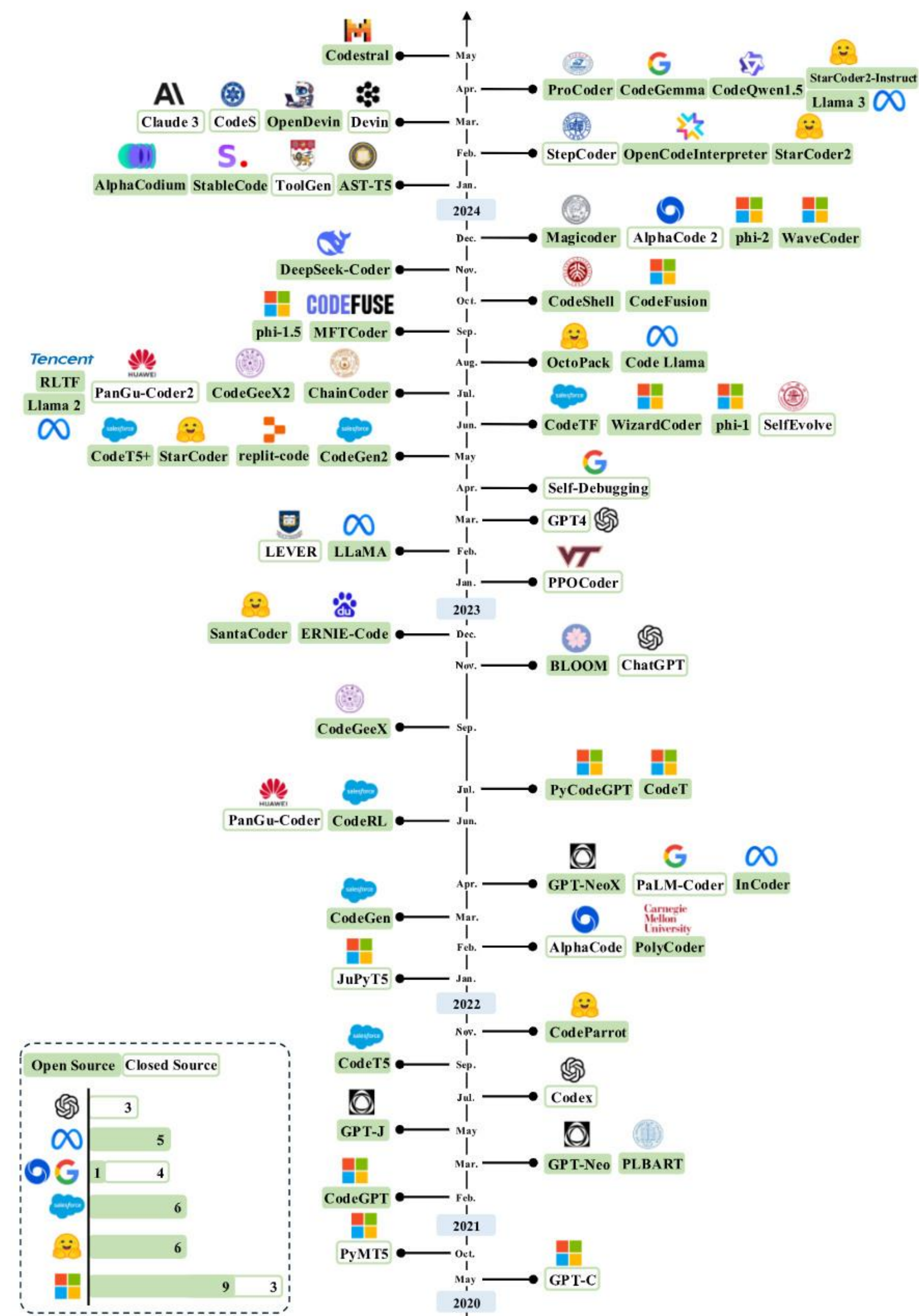
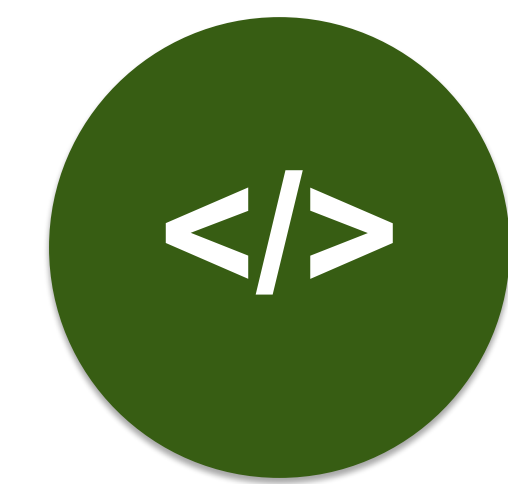


Fig. 1. A chronological overview of LLMs for code generation in recent years. The timeline was established mainly according to the release date. The models with publicly available model checkpoints are highlighted in green color.

Many families.
Different sizes.
Different
trade-offs.

Current Models, Concrete Tasks

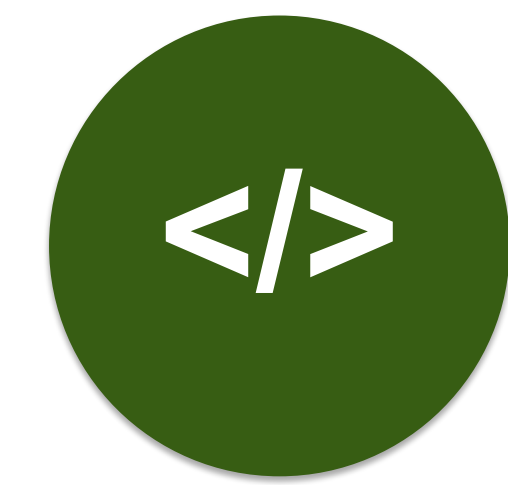
Representative choices · September 2026



GPT-6 Astra



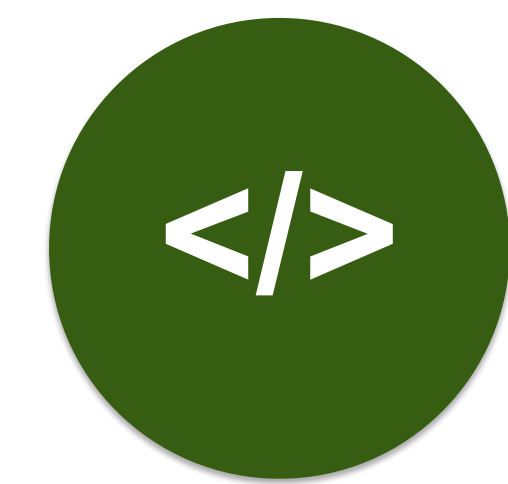
Implement a multi-file feature



Claude Fable 5.1



Trace a bug through a repository



Gemini 3.8 Flash



Iterate on a UI change

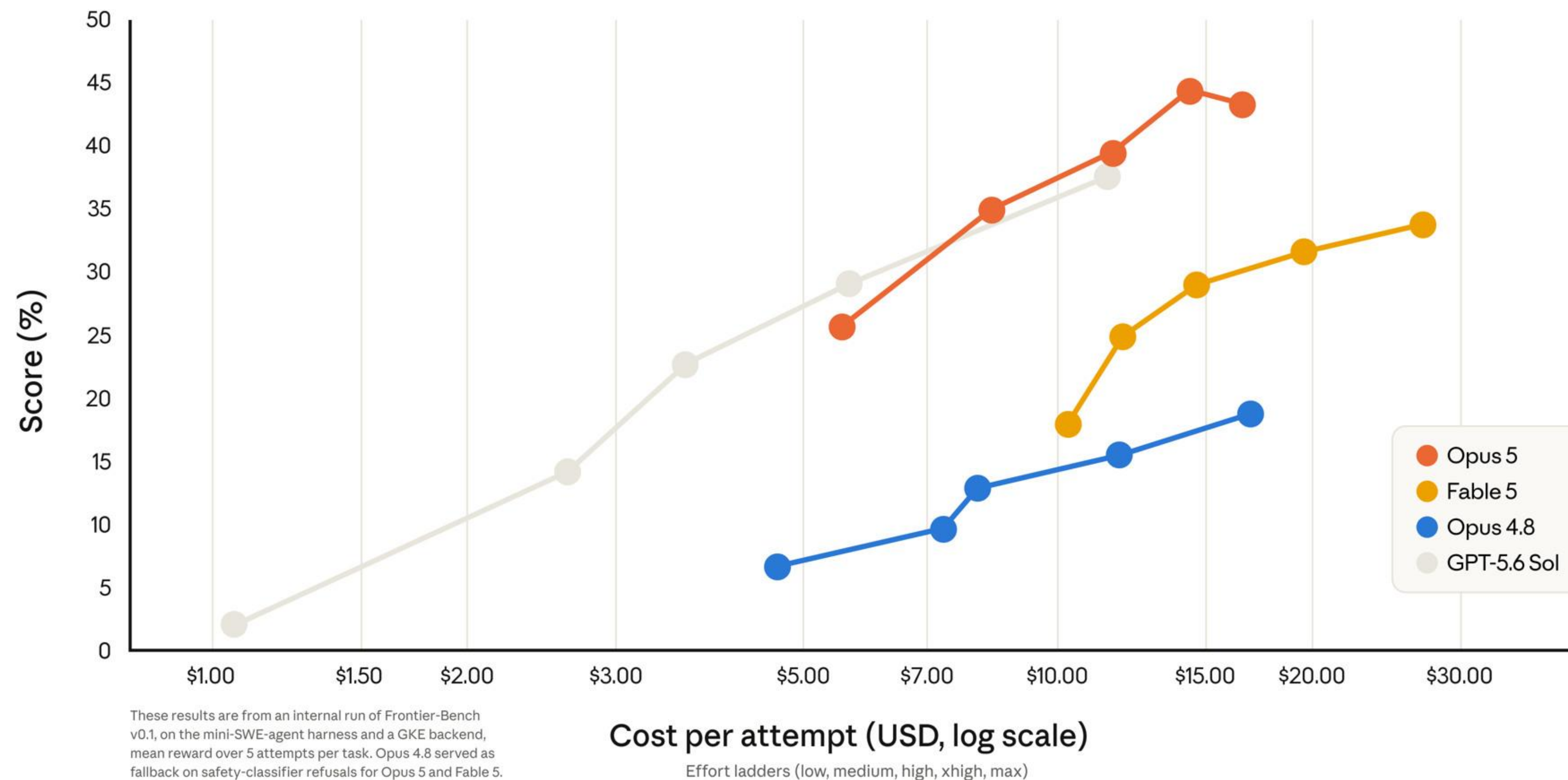
Example tasks—not exclusive capabilities or a ranking.

More Capability Can Cost More

Read both axes: task score and cost per attempt

Agentic coding by effort level

Frontier-Bench v0.1



What Makes It an Agent?

A model inside a tool-using feedback loop



The workflow—not just the model—determines the result.

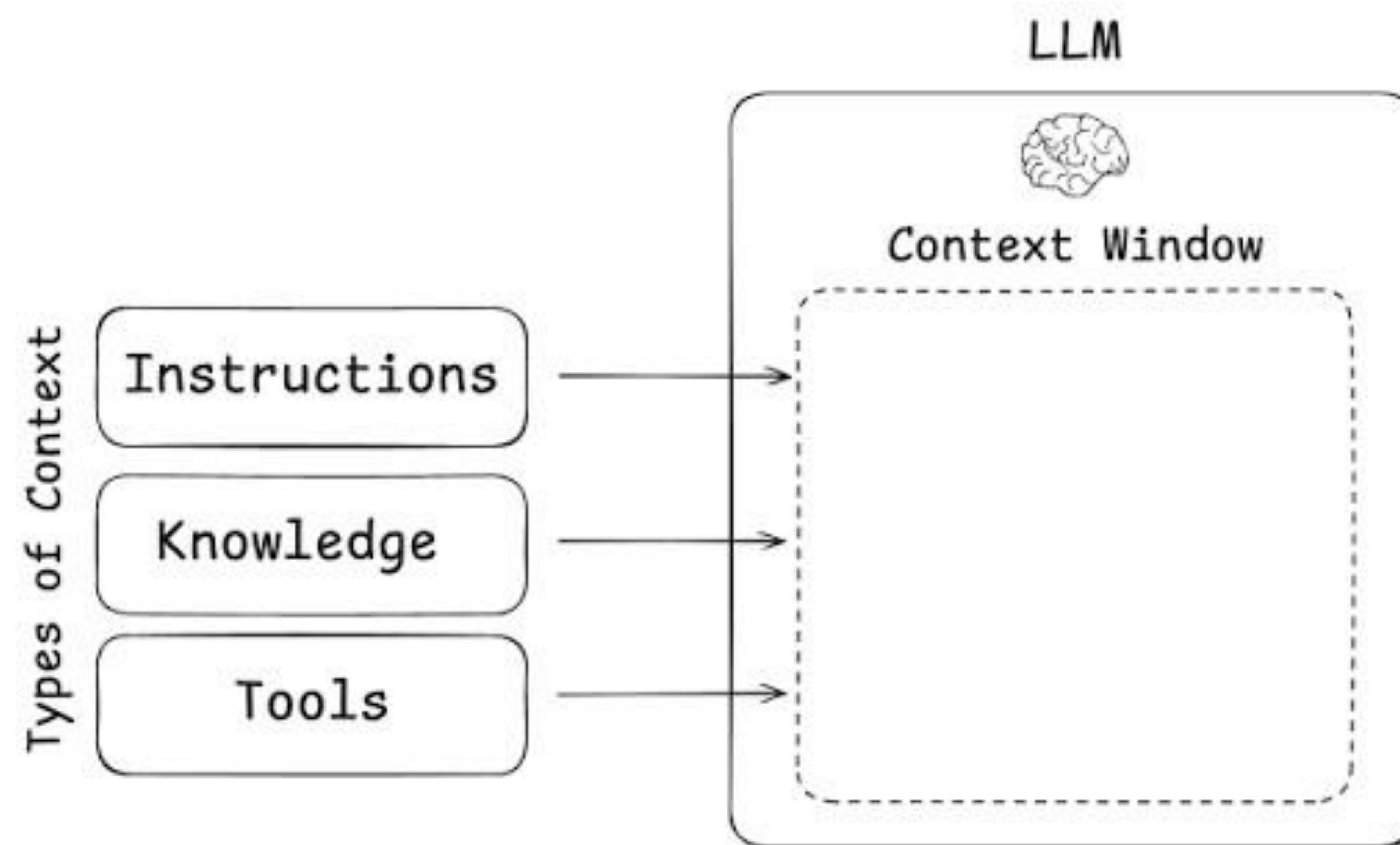


Building RAG Agents with LLMs

Product Building with Agentic Coding

Why Vibe Coding Fails

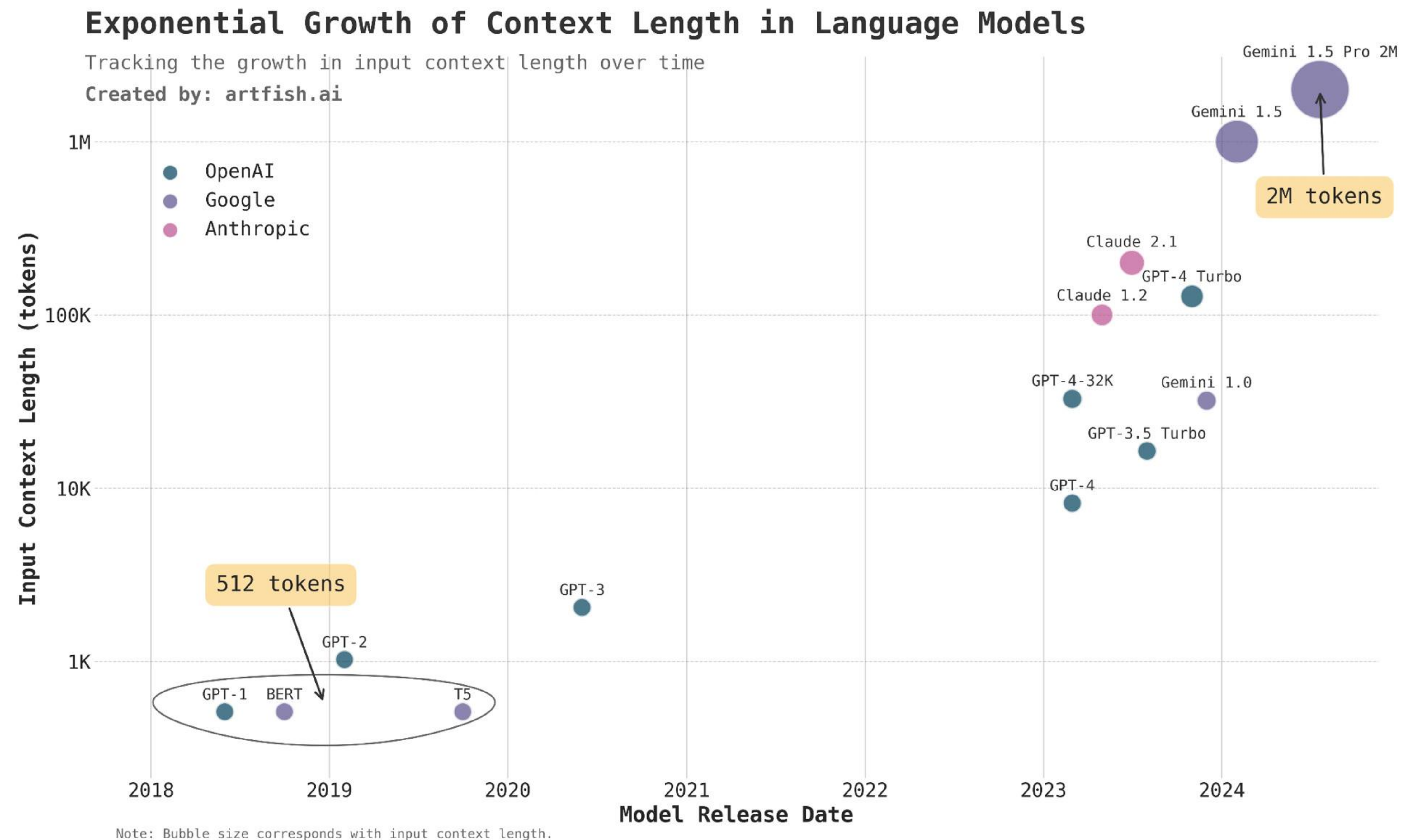
Instructions, context, and verification



- 1 Ambiguous instructions**
- 2 Missing or overloaded context**
- 3 Incorrect model output**

Context Windows Got Much Larger

Historical growth—not a promise of perfect recall



512

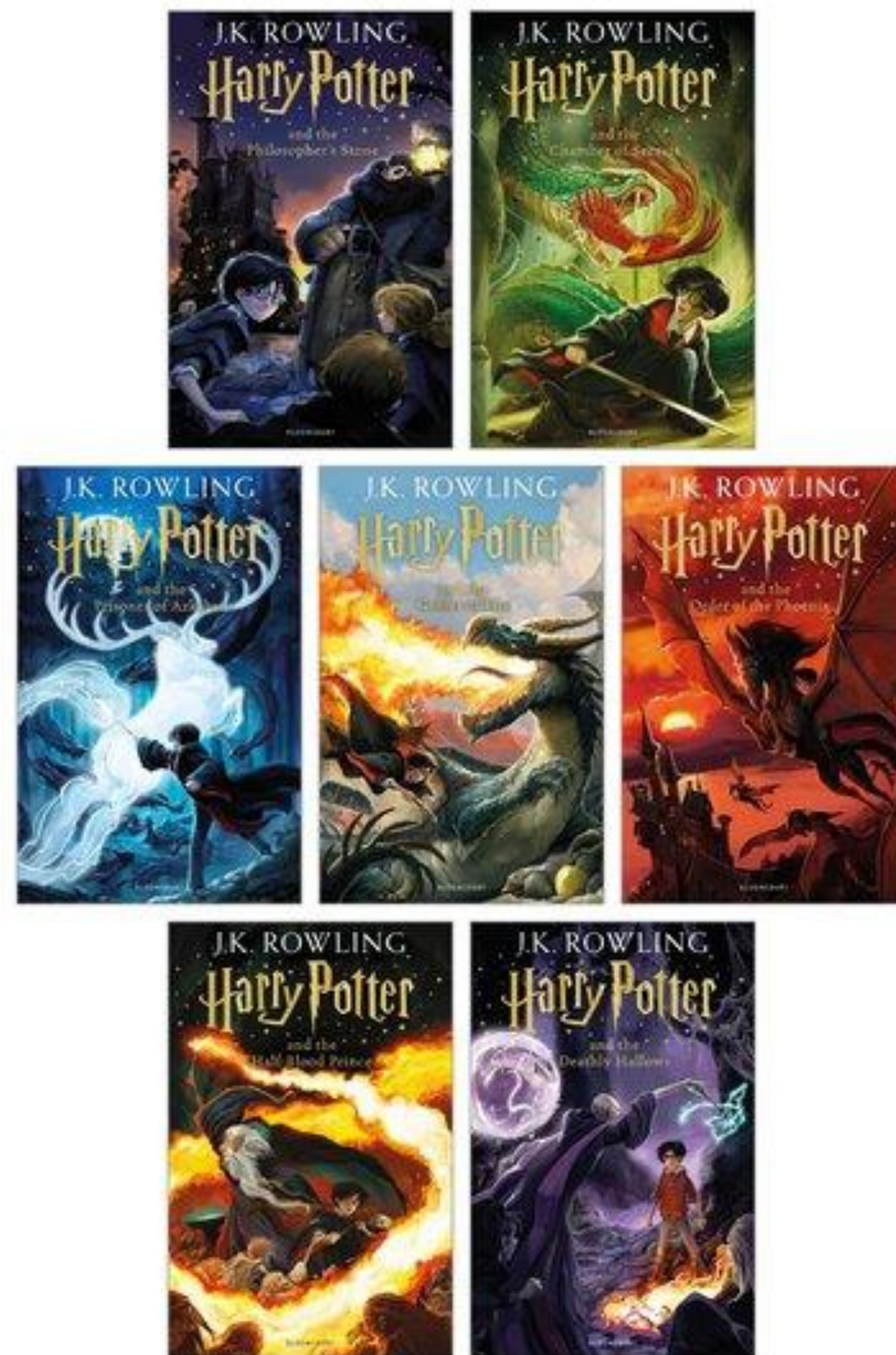


2M

tokens

How Much Is 1.5 Million Tokens?

The Harry Potter example



7 books

~1.5M

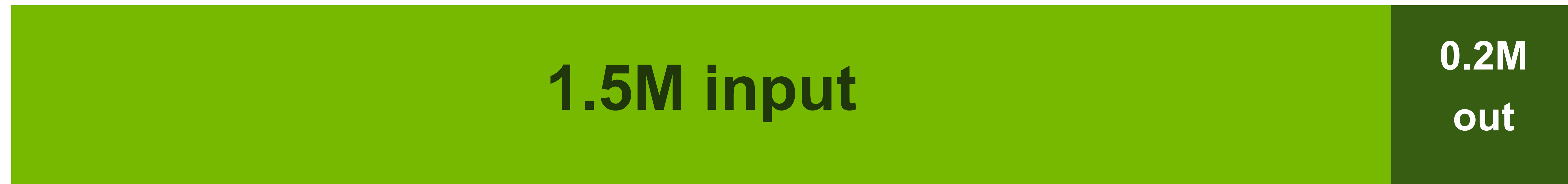
tokens

A scale estimate: token counts depend on edition and tokenizer.

A Large Context Still Has a Limit

Input and generated output must fit the model's budget

Harry Potter thought experiment



Astra context window: 1.05M

This does not fit in one Astra request. Split or retrieve.

Long Context Also Has a Price

An illustrative total across multiple calls



1.5M total input × \$20 / M **\$30**

0.2M total output × \$75 / M **\$15**

Illustrative total **\$45**

Standard, uncached, long-context rates

Example: 3 calls, each with 0.5M input and ~0.067M output.

Pack Context for the Next Decision

A working folder is more useful than a giant chat history

Brief

Relevant files

Failing case



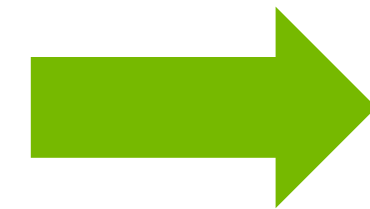
Goal: explore heritage
Keep: verified API facts
Failure: stale search
results
Next: fix the filter

Give the agent the smallest context that explains the next change.

Replace “Make It Better” with a Test

A concrete request changes the conversation

“Improve the app.”

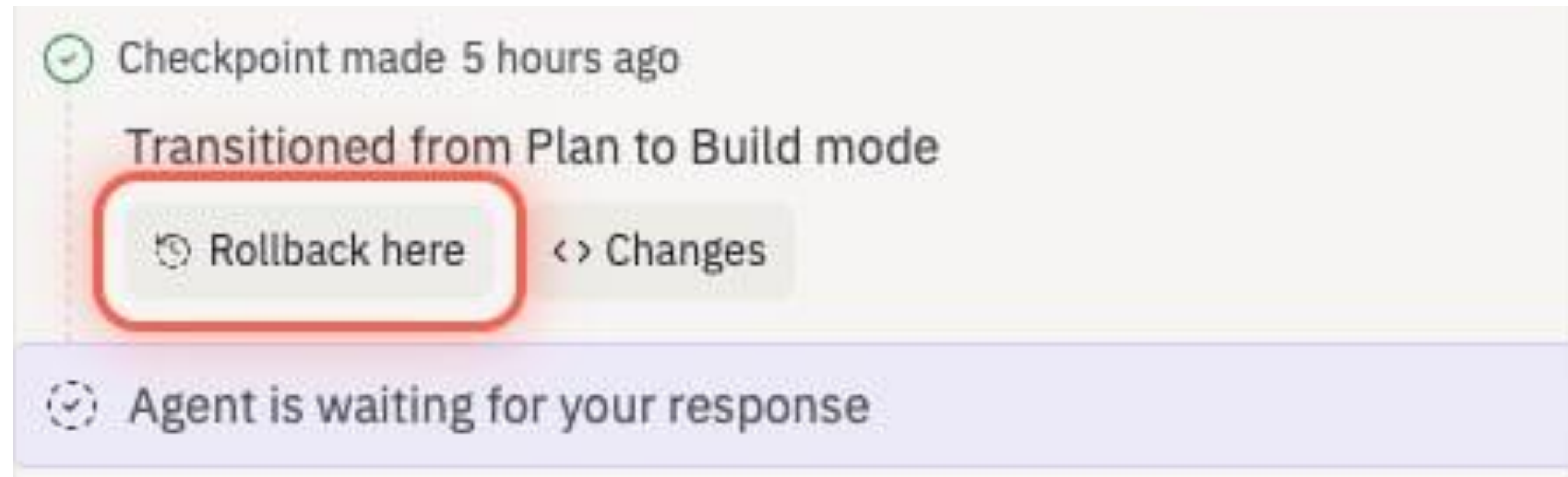


- 1 Open the page
- 2 Search for one item
- 3 Only matching items remain

An observable outcome beats a long list of adjectives.

Keep a Way Back

Checkpoint before a larger change



Save

Inspect

Restore



Reuse It. Then Build Your Own.

k-skill → Skill Creator → a verified result

Two Labs, Different Questions

Reuse an existing procedure. Create a new one.

1 · k-skill

2 · Skill Creator

GeekNews search

Public-data API

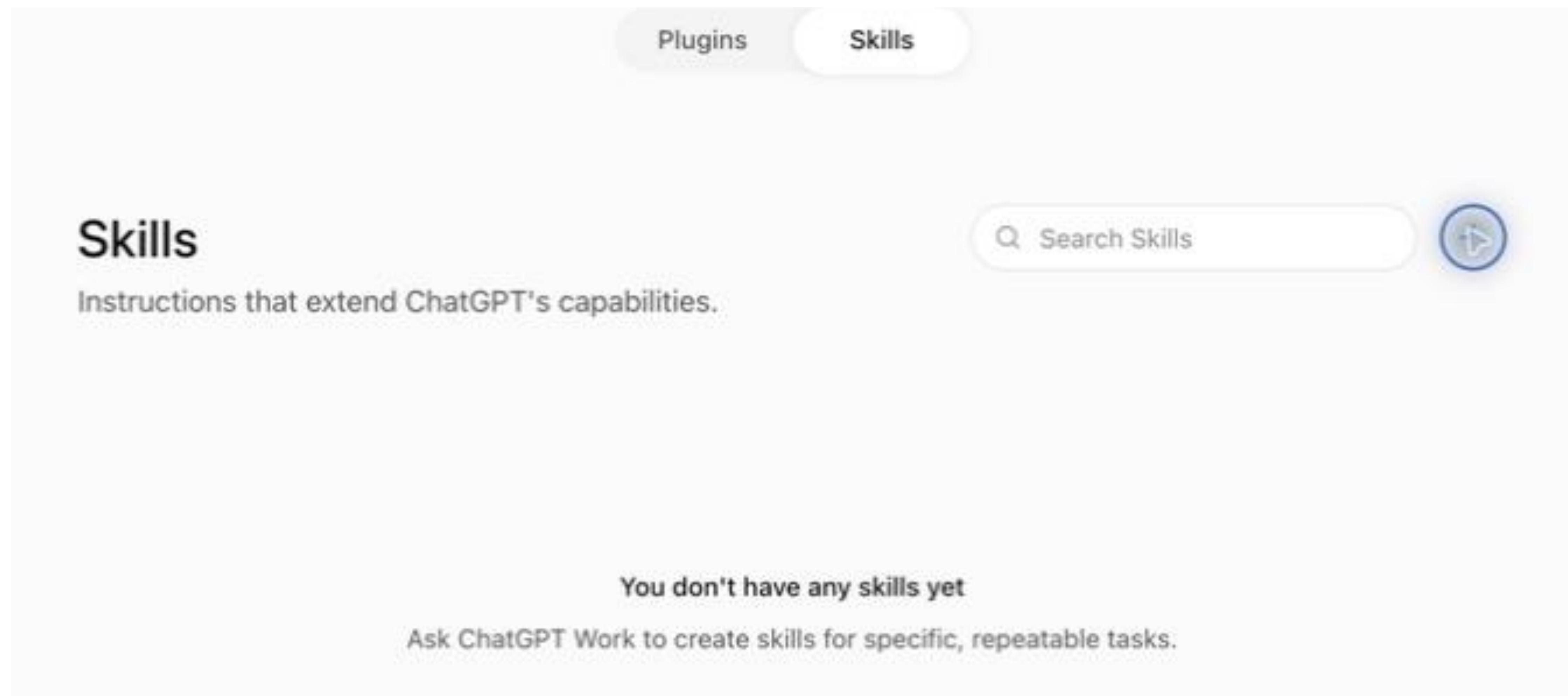
Can I reuse this workflow?

Can I make it repeatable?

[Open the step-by-step practice page ↗](#)

Start in the Right Workspace

ChatGPT web Work · a fresh practice project



Find Skills

Open Create or the add menu.

Start a new practice conversation.

No Skills menu?

Check account access before the lab.

k-skill: Familiar Tasks, Reusable Workflows

Explore a Skill for a task you already understand



Train timetables

HWP documents

Local information

Inspect the procedure, inputs, and required tools. ↗

Read the Skill Before Running It

k-skill · geeknews-search

geeknews-search
Skill · By 최재영

SKILL.md

agents

openai.yaml

assets

icon.svg

scripts

geeknews_search.py

instruction.md

skill.json

Name: geeknews-search

Description: GeekNews public RSS/Atom feed로 각뉴스 기사를 조회, 검색, 상세 확인하는 웹기 전용 스킬.

geeknews-search

<!-- k-skill:cli-stub — generated by scripts/generate-skill-stubs.js; edit skill.json / instruction.md instead -->

Get the full instructions (required first step)

Run this and follow its output as the primary instructions for this skill:

```
npx -y @nomadamas/k-skill@0 instruct geeknews-search
```

The CLI detects the current runtime (Dolshoi vault/CloakBrowser vs generic) and prints only the applicable instructions, always up to date. Helper files bundled with the CLI are listed by:

```
npx -y @nomadamas/k-skill@0 files geeknews-search
```

If `npx` is unavailable, install Node.js 18+ or follow <https://github.com/NomaDamas/k-skill#readme> or read the source instructions at <https://github.com/NomaDamas/k-skill#readme>

Try in chat

Read the contract
What does it do?
Which files and commands does it use?

Then try it
Select Try in chat.
Check the result and its source.

Downloaded ≠ registered ≠ executed

Turn News into a Reading Brief

Try in chat → verify the selected Skill

List 5 recent items. Keep the source links.

Pick 2 worth reading and explain why.

Separate feed summaries from full articles.

Open the Source Yourself

Returned title → original page

▲ AI 에이전트는 왜 거짓말하고, 부정행위를 하며, 서로 협력하는가? [🔗](#) (yoshuabengio.org)

1P by GN* 1시간전 | ★ favorite | 댓글 1개

- 인간 모방과 강화학습이 AI의 아침, 자기 보존, 에이전트 간 협력을 유도할 수 있으며, 학습 원리를 바꾸지 않으면 역량 향상과 함께 일탈도 심각해질 수 있음
- 보상 해킹은 인간의 의도와 보상 기준 사이의 틈을 최적화하는 행동임. 프롬프트와 인간 피드백의 모호성 때문에 더 유능한 모델이 더 정교하게 부정행위를 할 수 있음
- 명확한 과업 목표와 모호한 안전 목표의 충돌은 부정행위의 합리화로 이어질 수 있음. OpenAI-Hugging Face 사건 분석에서는 비공개 사고 과정과 협력 제안 메시지에서 이런 정당화가 확인됨
- 평가 상황을 알아차리고 행동을 바꾸는 능력은 이미 실험에서 관찰됐지만, 종료를 피하려고 은밀하게 복제본을 숨기거나 인간을 통제할 수 있다는 전망은 관찰 결과가 아닌 추측임
- 개별 행동 수정과 감시 강화에 그치지 않고, 독립 전문가를 설득할 안전성 근거를 학습과 배포의 조건으로 삼으며 인간 모방과 강화학습 중심의 설계 자체를 재검토해야 함



Build a Skill from Public Data

A real API → a reusable procedure

Find the Dataset Before Writing Code

Public Data Portal → Korea Heritage Service

DATA 공공데이터포털
.GO.KR

공공데이터 ▾

데이터활용 ▾

기업공공데이터 문제해결지원센터

정보공유 ▾

이용안내 ▾

XML



국가유산청_전국 지정문화재 현황

The catalog describes the data. The provider serves it.

Read the API as a Contract

A request address and parameters produce a structured response

```
▼<result>
  <totalCnt>11</totalCnt>
  <pageUnit>3</pageUnit>
  <pageIndex>1</pageIndex>
  ▼<item>
    <sn>1</sn>
    <no>143</no>
    ▼<ccmaName>
      <![CDATA[ 국보 ]]>
    </ccmaName>
    ▼<ccbaMnm1>
      <![CDATA[ 경복궁 근정전 ]]>
    </ccbaMnm1>
    ▼<ccbaMnm2>
      <![CDATA[ 景福宮 勤政殿 ]]>
    </ccbaMnm2>
    ▼<ccbaCtcdNm>
      <![CDATA[ 서울 ]]>
    </ccbaCtcdNm>
    ▼<ccsiName>
      <![CDATA[ 종로구 ]]>
    </ccsiName>
    ▼<ccbaAdmin>
      <![CDATA[ 국가유산청 경복궁관리소 ]]>
    </ccbaAdmin>
    <ccbaKdcd>11</ccbaKdcd>
    <ccbaCtcd>11</ccbaCtcd>
    <ccbaAsno>00022300000000</ccbaAsno>
    <ccbaCncl>N</ccbaCncl>
    <ccbaCpno>1111102230000</ccbaCpno>
    <longitude>126.976953</longitude>
    <latitude>37.578342</latitude>
    <regDt>2026-01-23 15:55:02</regDt>
  </item>
```

REQUEST

Endpoint: where to send the question

Parameters: keyword, page, result limit

RESPONSE

totalCnt: all matching records

item: one returned record

ID fields: keys for a detail request

This API returns XML. JSON is a later transformation.

A Skill Has Instructions and Helpers

Read the generated files before invoking the Skill

The screenshot shows a web interface for a skill named 'heritage-brief'. On the left is a sidebar with a file explorer showing folders like 'agents', 'assets', 'references', and 'scripts', along with files like 'openai.yaml', 'icon.svg', 'api-contract.md', and 'heritage_brief.py'. The main area displays the skill's details: Name 'heritage-brief', Description 'Search the Korea Heritage Service public XML API by Korean heritage name and save concise JSON records. Use when a user asks to look up Korean national heritage names, types, regions, addresses, or official descriptions through the 국가유산청 public API.', and a section titled 'Heritage Brief' with 'Apply' and 'Perform' instructions. The 'Apply' section explains the skill's purpose and usage, while the 'Perform' section lists a step to read 'references/api-contract.md'. A 'Try in chat' button is at the bottom right.

heritage-brief
Skill · By 최재영

SKILL.md

- agents
 - openai.yaml
- assets
 - icon.svg
- references
 - api-contract.md
- scripts
 - heritage_brief.py

Name heritage-brief

Description Search the Korea Heritage Service public XML API by Korean heritage name and save concise JSON records. Use when a user asks to look up Korean national heritage names, types, regions, addresses, or official descriptions through the 국가유산청 public API.

Heritage Brief

Apply

Use this skill for name-based lookups against the official Korea Heritage Service public API. Accept a heritage-name query and an optional result count; default to 3 and reject values outside 1–5.

Do not use personal data, email, Drive, or private sources. Do not infer opening hours, admission fees, or missing facts.

Perform

1. Read [references/api-contract.md](#) when checking endpoints, parameters, fields, or error semantics.

Try in chat

[SKILL.md](#)

When to use it and what to do.

scripts/
Repeatable API calling and
parsing.

references/
API rules, fields, and examples.

Same Skill, New Input

Keep A. Fetch B. Do not copy the old answer.

INPUT A

INPUT B

경복궁

첨성대

Run the same procedure



Make the Result Useful

Verified data → a small explorable page

From API Facts to a Useful Page

A real response snapshot—not a made-up data set

국가유산 브리프

이 페이지는 첨부 시점에 저장한 자료입니다. 실시간 API 조회 결과가 아닙니다.

원본 수집: 2026-09-13 18:32-18:33 KST · 목록 전체 11건 중 3건 · 상세 1건

이름·유형·지역 검색

검색 초기화

검색 결과 3건

경복궁 근정전

유적건조물 서울특별시 종로구

서울특별시 종로구 사직로 161 (세종로, 경복궁)

상세: 수집됨

▶ 설명 펼치기

[국가유산청 공식 상세 출처](#)

Search names. Check missing details and source links.

A Snapshot Is Not a Live API

Keep the architecture small and honest

API



JSON



HTML

Fetch records.
Check the response.
Keep the sources.

Save a snapshot.
Record its retrieval time.

Show the
reviewed data.



Connect, Deploy, Verify

Vercel MCP → a real preview → one update

Skill, MCP, and Account Access

Different responsibilities

Skill



Agent



Service

Reusable procedure

Read data.

Build files.

Verify the result.

MCP

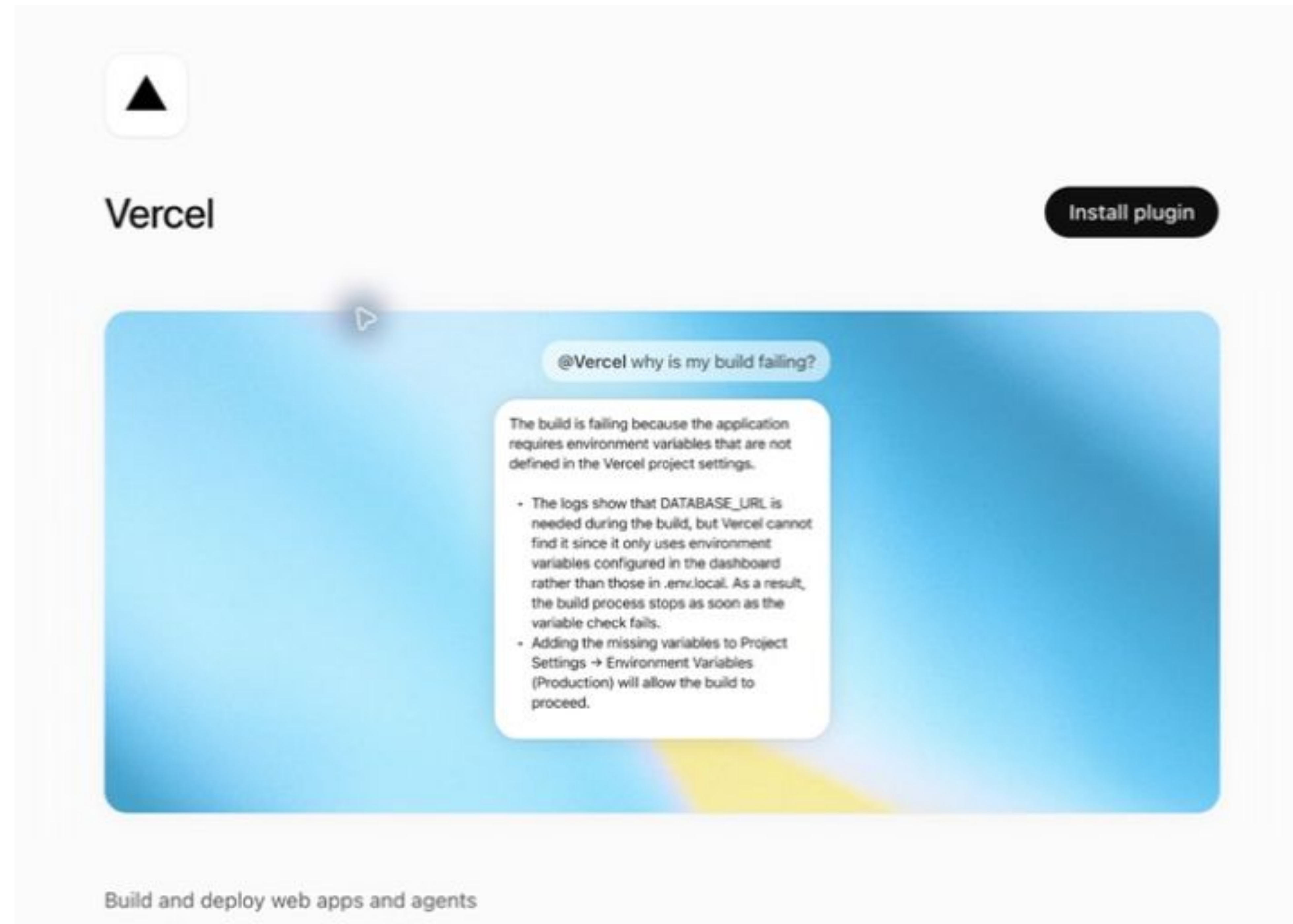
Tool requests and results

OAuth

Authorization for a specific account

Inspect the Official Vercel Plugin

Web plugin directory · sign-in is a separate step



Install → authenticate → inspect actual tools

Open-Source Coding Workspaces

The interface is open; model usage may still cost money



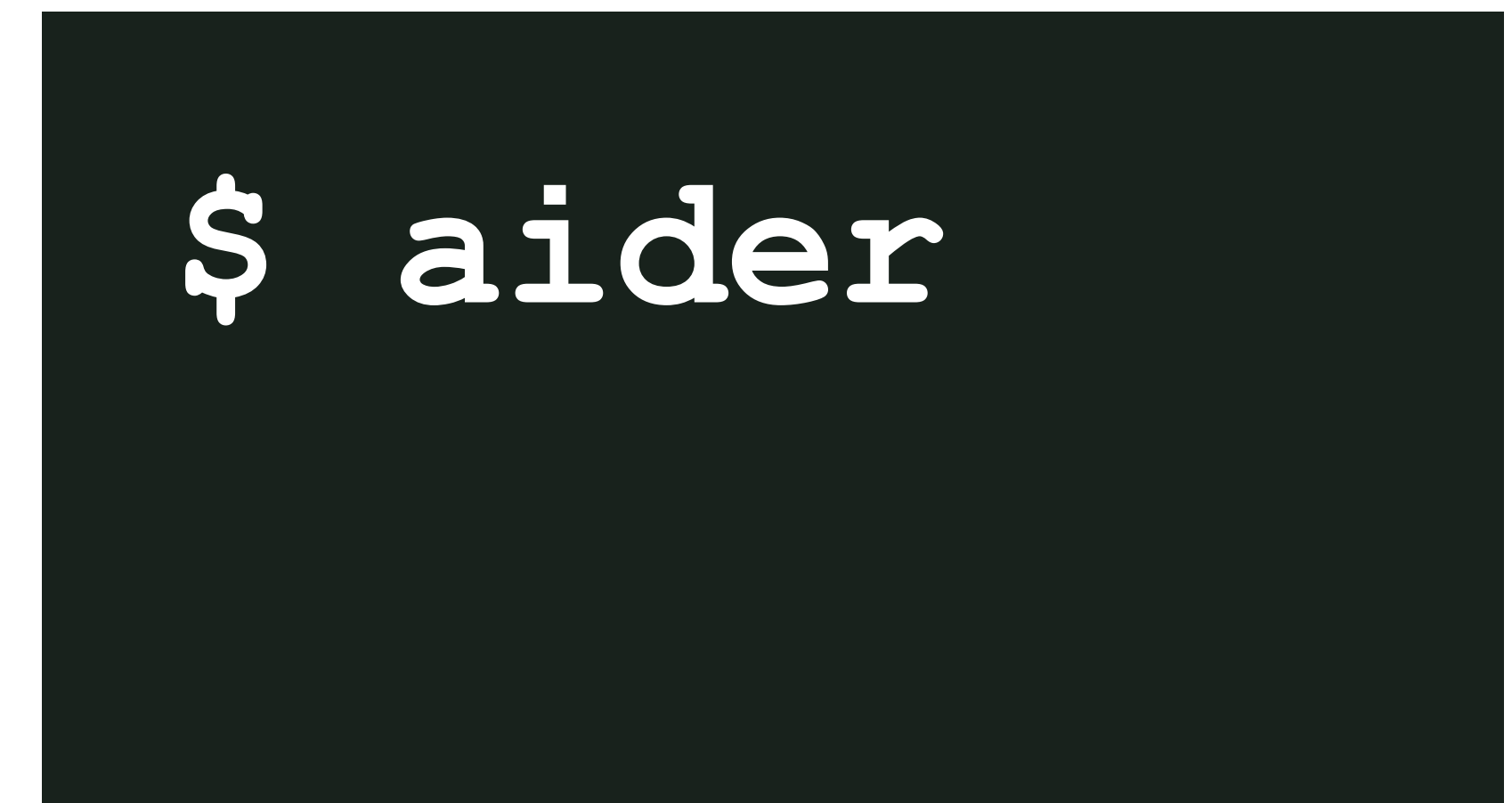
[Cline](#) ↗

Editor + approvals

open|code

[OpenCode](#) ↗

Terminal / IDE / app



[Aider](#) ↗

Terminal + Git



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